TRISTAN GABL

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EDUCATION

Computational Science and Engineering at ETH Zürich is a comprehensive interdisciplinary program combining applied mathematics, computer science, and engineering, that can be focused on robotics, machine learning, and numerical methods.

ETH Zürich *2025 — present*
Zurich, Switzerland
Master of Science ETH in Computational Science and Engineering **GPA 5.35/6 (current)**

- Robot Dynamics
- Robot Learning
- Probabilistic Artificial Intelligence
- Advanced Numerical Methods
- Recursive Estimation
- Model Predictive Control
- Large-Scale AI Engineering

ETH Zürich *2021 — 2025*
Zurich, Switzerland
Bachelor of Science ETH in Computational Science and Engineering **GPA 5.01/6**

- Computer Vision
- 3D Vision

EXPERIENCE

Module Lead for Perception, AMZ Racing, Driverless *2025 — 2026*
Zurich, Switzerland

- Lead the next generation of team members responsible for all perception related subsystems.
- Planned and led weekly meetings for timeline management and to ensure knowledge transfer.
- Ensured coherence of development goals across the perception, estimation, and controls modules.

Core Member for Sensor Fusion, Perception, AMZ Racing, Driverless *2024 — 2025*
Zurich, Switzerland

- Developed bird's eye view sensor fusion model (ResNet-style backbone with CenterNet-style detection head), combining LiDAR, camera, and odometry data for landmark detection at high-speed driving (20 m/s) within latency budget, deployed as a ROS2 inference node.
- Built a custom Python dataset pipeline extracting past-season rosbags and transforming globally recorded data into the new car's local sensor frames for training, with data augmentations for model robustness.
- Improved and maintained performance of baseline detection methods (LiDAR only, camera only) across incoming software and hardware changes of other modules and the car.
- Profiled and optimised the inference pipeline to stay within the 50 ms latency constraint, and maintained a clean, modular, well-documented codebase for efficient knowledge transfer.

Research Assistant at Computer Vision and Geometry Group, ETH Zürich *2026*
Zurich, Switzerland

- Collecting and pre-processing camera, Aria glasses and LiDAR data for a computer vision research project about human interactions in a kitchen setting.

Teaching Assistant at ETH Zürich *2023 — present*
Zurich, Switzerland

- Teaching assistant for Computer Science 1 & 2 across several semesters.

Paramedic at Red Cross at Civil Service *2020 — 2021*
Austria

- Gained experience in taking responsibility in critical situations.

PROJECTS

Cloth folding using a diffusion policy on a SO-101 **(github)**
Robot Learning

- Established a data collection workflow for imitation learning, used to train and verify diffusion policies.
- Refactored the codebase to add support for Meta's DINOv3 vision backbone and a lighter flow matching model without the text encoder, matching or outperforming the CNN-based diffusion policy with 1/5 of the parameters and enabling rollouts on a standard M1 Mac.
- Generalised the policy to arbitrary towel colours by segmenting the towel in every recorded frame with SAM 2 and adding a colour-changing data transform to LeRobot, combined with a dataset of varied offsets and rotations.

CuSSS: Cuda implementation of SSS variants **(github)**
Large-Scale AI Engineering — Awarded a letter of distinction
- Implemented native Cuda kernels for variants of SSS activation functions, focused on runtime.
- Validated kernels against PyTorch autograd equivalents and verified speedups using benchmarks and Nsight Systems profiling.

Action recognition using 3D hand-object contact maps **(github)**
3D Vision
- Predicted performed actions from short sequences of hand and object vertices on the egocentric H2O dataset.
- Compared two-stage and end-to-end training of MLP and transformer models, using thresholded contact and distance maps as an intermediate representation.

Software pipeline for 3D scene graphs **(github)**
Bachelor's thesis
- Developed code for combining point clouds, 2D panoptic segmentation, and MLP output to produce 3D scene graphs.
- Separated object instances on the mesh using connectivity, best-view re-projection of Mask2Former instances for duplicate removal, and BFS assignment of leftover vertices; evaluated with semantic and instance mIoU on ScanNet scenes.
- Compiled a relationship dataset from 3RScan/3DSSG and trained an embedding-based MLP to predict relations between objects; results shown in an interactive Plotly viewer.

SKILLS

Languages: English • German
Technologies: C • C++ • Python • PyTorch • ROS2 • Cuda • Julia • MATLAB • JavaScript • Git • Linux • Final Cut Pro
Interests: Robot learning • Autonomous systems • Hardware involved projects